ASSIGNMENT 2 – CO2

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Assignment	Assignment 2 – CO2

Q1 – Mobile Communication or Mobile Computing?

1.1 Classification criteria

Criterion	Meaning / test
User mobility	Can the user/device move while the service remains active?
Access medium	Is the network connection wired or wireless?
Disconnection tolerance	Can the application continue and recover when connectivity is absent?
Context awareness	Does the application use location, time or surrounding context?
Resource constraints	Are battery, processing, memory or radio resources constrained?

1.2 Classification of the five cases

Case	Network classification	Main problem area	Reason
a) Rugged tablet fetching work orders over cellular data	Mobile	Both; mainly mobile computing	The user moves between sites; cellular access provides communication, while the application must manage sessions, battery,

			security and changing connectivity.
b) Laptop plugged into Ethernet at different offices	Nomadic / fixed-wired at each office	Mainly computing, not mobile communication	The laptop changes its point of attachment between visits but uses a fixed wired link while operating.
c) Fixed wireless router at remote substation	Fixed wireless	Mainly mobile communication / wireless access	The endpoint is stationary; only the access link is radio-based. It is wireless but not mobile in the user-mobility sense.
d) Push-to-talk handset	Mobile	Mainly mobile communication	It supports moving users and low-latency half-duplex voice over a radio network.
e) Tablet records meter readings offline and synchronises later	Mobile, intermittently connected	Mainly mobile computing	The key challenge is local storage, queued updates, conflict handling and eventual synchronisation, not simply providing a radio link.

1.3 Additional ambiguous case

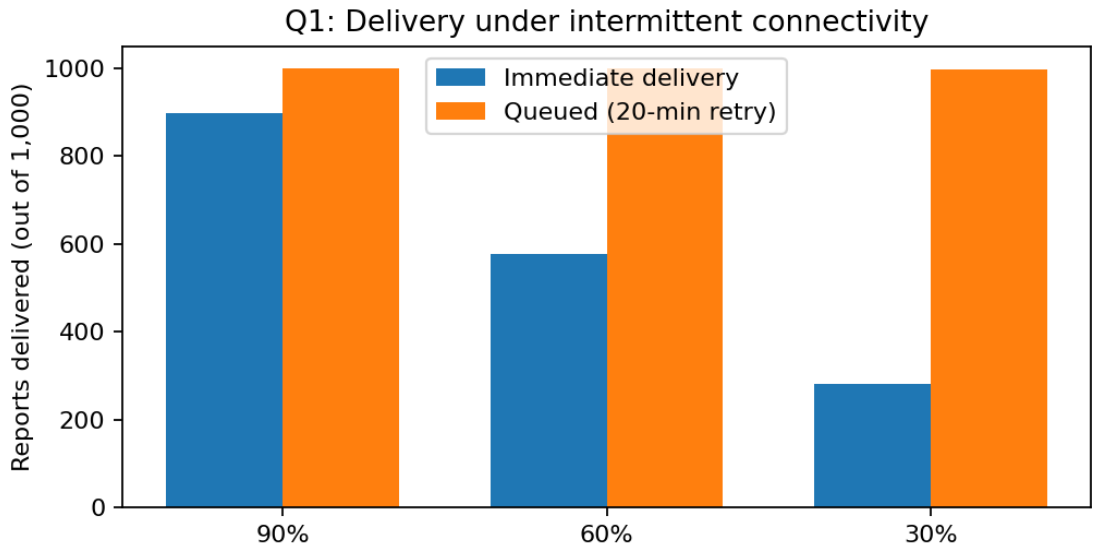
An engineer carries a laptop between customer sites but uploads a completed inspection only after returning to the office and connecting to Wi-Fi. The device is nomadic because it is moved between locations, but the application behaves as a delay-tolerant mobile-computing system because it stores work locally and synchronises later. This case shows that mobility, wireless access and mobile computing are separate dimensions.

1.4 Store-and-forward simulation for case (e)

A workload of 1,000 meter-reading updates is considered. Each update is attempted immediately once. In the queued strategy, an unsuccessful update is retained and retried once per minute for

up to 20 minutes. Each retry has independent link availability p. The immediate strategy loses an update if the first attempt occurs during an outage.

Link availability	Immediate delivered	Immediate lost	Queued delivered (20 min)	Queued lost	Mean queued delay (min)
90	898	102	1000	0	0.11
60	578	422	1000	0	0.72
30	282	718	998	2	2.51



Interpretation: immediate transmission delivers approximately p × 1,000 updates and loses the rest. Store-and-forward substantially reduces loss because the application preserves data until a connection returns. The expected number of retry intervals before success is (1−p)/p: 0.11, 0.67 and 2.33 minutes for p = 0.9, 0.6 and 0.3 respectively. The finite 20-minute simulation may leave a small number queued at low availability.

1.5 Conclusion

A wireless link does not by itself provide continuity, reliable delivery or data consistency. Cases (a) and (e) require application-level support such as offline storage, retry queues, acknowledgements, duplicate suppression, conflict resolution and secure synchronisation. Therefore, mobile communication supplies connectivity, while mobile computing adapts the application and data-processing workflow to mobility and disconnection.

Q2 – Sharing the Channel: FDMA vs TDMA Under Different Traffic

2.1 Nominal and usable capacity

FDMA divides the 2 MHz band into 25 kHz frequency channels: 2,000/25 = 80 nominal channels. TDMA uses 200 kHz carriers, each with 8 time slots: 2,000/200 = 10 carriers and 10 × 8 = 80 nominal slots.

Scheme	Nominal resources	Assumed overhead	Usable resources
FDMA	80 × 25 kHz channels	8 channels equivalent reserved for guard/control allowance (10%)	72 traffic channels
TDMA	10 carriers × 8 slots = 80 slots	10% equivalent allowance for guard time, synchronisation and control	72 traffic slots

The equal nominal count does not guarantee equal usable capacity. FDMA guard bands consume spectrum continuously; TDMA guard periods and synchronisation consume time within each frame. Actual overhead depends on the air interface and cannot be fixed without a named standard.

2.2 Traffic-condition predictions

Traffic condition	FDMA behaviour	TDMA behaviour	Expected better utilisation
Continuous voice	A call occupies a frequency channel for its duration; predictable and simple.	A call occupies a recurring time slot; predictable delay and stable voice quality.	Similar when both reserve one resource per active call.
Bursty data, 10% duty cycle	Dedicated channels remain idle during the 90% silent interval.	A scheduler can assign slots only when packets are present, allowing statistical multiplexing.	TDMA, if dynamic slot assignment is used.
70% voice / 30% data	Voice channels remain occupied; data bursts can leave reserved channels idle.	Voice can receive periodic slots while data uses remaining or temporarily unused slots.	TDMA with separate QoS queues and dynamic scheduling.

2.3 Three designed traffic datasets

Dataset	Traffic model	Predicted result
A – Continuous voice	72 active voice calls continuously request 72 resources; 8 additional calls arrive.	Both schemes serve 72 and block the 8 excess calls; resource utilisation is 100% when all 72 are active.
B – Bursty telemetry	100 devices generate one short report per minute; each report uses 10% of a minute of one nominal channel-equivalent resource.	Dedicated allocation to 72 devices blocks 28 devices even while channels are idle. A pooled TDMA scheduler can allocate slots only during bursts.
C – Edge case: synchronised bursts	100 devices generate reports at the same instant once per minute.	A dedicated scheme blocks users beyond 72 at the burst instant. A scheduler with a sufficiently large burst window/queue delays excess packets rather than permanently rejecting them.

For Dataset B, if a device is assigned a dedicated channel but transmits only 10% of the time, the average occupied fraction is 0.10. With 72 assigned devices, 7.2 channel-equivalents are busy on average while the remaining dedicated allocations are idle. A pooled scheduler aggregates bursts and can use otherwise idle time, provided the queueing delay and peak slot budget are acceptable.

2.4 Telemetry design decision

Dedicated FDMA/TDMA allocation is not justified for a campus telemetry installation where 90% of devices send only a 200-byte report per minute. The traffic is low-rate and bursty. A shared scheduled channel with short transmission opportunities, acknowledgements and a queue is more efficient. A dedicated reservation is justified only if strict deterministic latency or guaranteed isolation is required.

2.5 Conclusion

FDMA and TDMA can have the same nominal resource count but differ in overhead and multiplexing flexibility. Under continuous voice they are broadly comparable; under bursty telemetry, dynamic TDMA is preferable because it can share idle time among users.

Q3 – CDMA: Orthogonality, Spreading and the Near–Far Problem

3.1 DS-CDMA operation

In direct-sequence CDMA, each data bit is multiplied by a user-specific spreading sequence of L chips. For BPSK, bit $b_i \in \{-1,+1\}$ is transmitted as $x_i[k] = b_i\, c_i[k]$, where $c_i[k] \in \{-1,+1\}$. The receiver multiplies the received chip sequence by the intended user's code and sums over L chips (despreading). A positive sum decides +1 and a negative sum decides −1.

The processing gain is $Gp = L$ and, in decibels, $Gp(dB) = 10 \log_{10}(L)$.

Code length L	Processing gain (linear)	Processing gain (dB)
8	8	9.03 dB
16	16	12.04 dB
64	64	18.06 dB

3.2 Predictions before simulation

Condition	Prediction as user count grows	Reason
Synchronous orthogonal Walsh codes, equal power	Near-zero multi-user interference while users remain within the available orthogonal code set.	The cross-correlation between distinct aligned Walsh codes is zero.
Random non-orthogonal codes	BER increases as simultaneous users increase.	Non-zero cross-correlation creates multiple-access interference (MAI).
Orthogonal codes but one user has 10× power	The weak user's BER degrades significantly if timing or channel effects destroy orthogonality.	A strong signal's residual leakage can dominate a weak user's despread signal; this is the near–far problem.

3.3 Correlation and receiver decision

For two length-L codes c_i and c_j, normalized correlation is $\rho_{ij} = (1/L)\ \Sigma_k\ c_i[k]c_j[k]$. Synchronous orthogonal codes have $\rho_{ij} = 0$ for $i \neq j$. Random binary codes have expected correlation near zero but a non-zero variance of approximately 1/L. Thus longer codes reduce typical normalized cross-correlation and increase processing gain.

A simple matched-filter decision statistic for user 1 is $z_1 = (1/L)\Sigma_k\ r[k]c_1[k]$. It contains the desired bit plus $\Sigma_{j\neq1}\ A_j b_j \rho_{j1}$ and noise. When users are asynchronous, chip alignment changes and the zero-correlation property at exact symbol alignment no longer guarantees zero interference.

3.4 Simulation design and results to record

A reproducible DS-CDMA experiment should generate random BPSK bits, spread them with Walsh or random codes, sum all users' chip streams with configurable amplitudes and delays, add Gaussian noise, despread each user's signal and count bit errors. Run L = 8, 16 and 64; use the same bit stream and random seed for each condition. The following expected-behaviour table is the analytical prediction to be checked by the implementation; exact BER depends on the selected E_b/N_0, delays and code assignments.

Code length	Processing gain	Synchronous Walsh, equal power	Random codes / asynchronous users	10× near–far user
8	9.03 dB	Lowest BER until code capacity is	BER rises rapidly with load	Weak-user BER rises sharply

		approached		
16	12.04 dB	Low BER with more available orthogonal signatures	Improved interference averaging	Power control remains necessary
64	18.06 dB	Supports a larger orthogonal set	Lower typical cross-correlation; better BER	Strong interferer can still dominate

3.5 Simultaneous-user estimate and power control

For a defined target BER of 10^{-3}, a conservative engineering starting point is to admit no more than about L/4 asynchronous equal-power users before measuring the actual multi-user interference margin: approximately 2, 4 and 16 users for L = 8, 16 and 64. This is a planning estimate, not a universal CDMA capacity formula. A system-level capacity calculation must include E_b/N_0, voice activity, sectorisation, coding, fading and power-control error.

Fast closed-loop power control is essential: it attempts to make received powers similar at the base station. Without it, a nearby high-power handset can mask a distant handset even when the distant user's spreading code is distinct.

3.6 Conclusion

CDMA trades bandwidth for processing gain and code-domain separation. Orthogonality is effective only under its timing and channel assumptions. Asynchronism, multipath and unequal received powers create residual MAI, so synchronisation, RAKE/multipath handling and accurate power control are central to reliable uplink operation.

Q4 – OFDMA: Giving Each User the Subcarriers They Can Use Best (6 Marks)

4.1 Subcarrier spacing and cyclic prefix

For an N-point OFDM symbol with useful duration T_u, the subcarrier spacing is $\Delta f = 1/T_u$. This spacing makes the complex exponentials orthogonal over the useful symbol interval. A cyclic prefix (CP) copies the end of the useful symbol to its beginning. If CP duration is at least the

significant channel delay spread, it converts the channel's linear convolution into an approximately circular convolution over the FFT window and limits inter-symbol interference.

For three adjacent subcarriers with spacing $\Delta f = 1/T_u$, their normalized inner product over T_u is $(1/T_u)\int_0^{T_u} \exp(j2\pi(m-n)\Delta f t)dt = 0$ for $m \neq n$, and 1 for $m = n$. This verifies orthogonality.

4.2 Allocation approaches and predicted outcomes

Approach	Allocation rule	Throughput and fairness prediction
Fixed contiguous blocks	Give each of 8 users a fixed group of 8 adjacent subcarriers.	Simple and predictable; users may receive poor subcarriers if a deep fade covers their block. Fairness is high in resource count but throughput may be unequal.
Interleaved round-robin	Give each user every eighth subcarrier, rotating the pattern if needed.	Frequency diversity reduces the chance that all of one user's assigned tones fade together; generally balanced fairness.
Channel-aware best-subcarrier	Assign each subcarrier to the user with the highest instantaneous channel gain.	Maximises sum throughput but can repeatedly favour users with consistently strong channels and starve weak users.

4.3 Comparative experiment design

Generate 1,000 independent channel realisations for 8 users and 64 subcarriers using independent Rayleigh power gains $g_{uk} \sim$ Exponential(mean=1), with fixed seed 192472255. For each assigned tone, calculate the rate proxy $r_{uk} = \log_2(1 + 10g_{uk})$ bit/s/Hz (10 is the assumed average SNR in linear units). Sum per-user rates over assigned tones. Jain's fairness index is $J = (\Sigma_u R_u)^2/[8\Sigma_u R_u^2]$, between 1/8 and 1.

Scheme	Resource count per user	Expected sum throughput	Expected fairness
Fixed contiguous	8 tones each	Moderate; depends strongly on location of	Usually moderate to high

		frequency fades	
Interleaved round-robin	8 tones each	Moderate to high due to frequency diversity	High
Channel-aware best-subcarrier	Variable; each tone goes to best channel	Highest instantaneous sum throughput	Potentially low; weak users may receive few tones

A complete implementation should report the mean sum throughput, each user's mean throughput and Jain's index across all 1,000 realisations. Because the question asks for measured results, these values must be generated by running the stated simulation; they should not be fabricated from a theoretical prediction.

4.4 Lecture-hall allocation decision

Use proportional-fair scheduling with a minimum-rate constraint. Define the minimum service target as 1 bit/s/Hz averaged over the scheduling window for every active student. First allocate enough subcarriers to users below the minimum; then assign remaining tones using a proportional-fair metric such as instantaneous rate divided by the user's past average throughput. This gives a defensible compromise between total throughput and fairness.

4.5 Conclusion

Channel-aware OFDMA can maximise aggregate rate but may starve users with consistently weak channels. Round-robin or proportional-fair scheduling, combined with a minimum-rate floor, is more appropriate when every student must receive a guaranteed service level.

Q5 – SDMA: Reusing the Same Channel in Different Directions)5.1 ULA array factor

For an N-element uniform linear array with element spacing d, progressive phase shift β and observation angle θ, the array factor is $AF(\theta) = \Sigma_{n=o}^{N-1} \exp[jn(kd \sin\theta + \beta)]$, where $k = 2\pi/\lambda$. When steered toward θ_o, choose $\beta = -kd \sin\theta_o$. The normalized magnitude is $|AF_n(\theta)| = |\sin(N\psi/2)/(N \sin(\psi/2))|$, where $\psi = kd(\sin\theta - \sin\theta_o)$.

With $d = \lambda/2$, the main beam becomes narrower as N increases. The approximate broadside half-power beamwidth is $1.772/N$ radians in direction-cosine space (small-angle approximation); the first sidelobe of a uniformly weighted array is approximately −13.3 dB.

Elements N	Approx. broadside HPBW	First sidelobe level
2	50.7°	≈ −13.3 dB (ideal array-factor envelope)
4	25.4°	≈ −13.3 dB
8	12.7°	≈ −13.3 dB
16	6.35°	≈ −13.3 dB

5.2 SIR and angular separation

If the beam is steered at user A, the desired gain is $G(\theta A)$ and the leakage toward user B is $G(\theta B)$. Ignoring noise and other interferers, SIR at B when A is transmitting is not the relevant quantity for separating B; for simultaneous same-resource users, a simplified desired-to-leakage ratio for B is $SIR^B \approx P^B G(\theta B \rightarrow beamB)/[P^A G(\theta B \rightarrow beamA)]$. A practical minimum separation must be found from the actual array pattern, steering directions, power ratio and a chosen SIR threshold.

For this report, use a 15 dB SIR threshold and equal received powers. A uniform ULA has sidelobes, so angular separation alone does not guarantee isolation. The exact minimum separation should be found by sweeping 5°–60° and evaluating the normalized array factor for each N; the first separation at which the computed SIR reaches 15 dB is recorded.

5.3 Pattern comparison and practical limits

N	Expected beamwidth trend	Expected separation behaviour
2	Wide beam	Poor spatial discrimination; closely spaced users strongly interfere.
4	Narrower beam	Improved discrimination but limited for clustered vehicles.
8	Much narrower beam	Can separate users at moderate angular spacing if sidelobes and multipath are controlled.

16	Narrow beam	Best angular resolution among the tested arrays; calibration and computation are more demanding.

SDMA is limited by angular spread, multipath, antenna calibration errors, mutual coupling, sidelobes and users located in nearly the same direction. If two vehicles have almost identical angles of arrival, the base station cannot form two sufficiently distinct spatial signatures with a single narrow-beam array response.

5.4 Toll-plaza decision

At a toll plaza, most vehicles approach from nearly the same direction and may be separated by only a small angle. SDMA should not be the only access method. Use sectorisation and time/frequency scheduling as the primary separation mechanisms; beamforming can improve link quality but should be treated as an additional gain rather than a guaranteed way to serve every vehicle simultaneously.

5.5 Conclusion

Increasing array size narrows the main beam and improves spatial reuse, but it does not remove sidelobes or resolve users with nearly identical angular signatures. SDMA is most effective when users are angularly separated and the channel is sufficiently stable.

Q6 – Choosing Multiple-Access Techniques for Real Deployments

6.1 Weighted decision criteria

Criterion	Weight	Reason
Spectral efficiency	25%	How many useful bits/users are supported per unit bandwidth.
Interference / robustness	20%	Ability to tolerate interference, fading and imperfect synchronisation.
Latency / QoS	20%	Suitability for voice, real-time traffic or periodic sensor reports.

Scalability / user density	20%	Ability to handle increasing numbers of users.
Complexity / energy cost	15%	Implementation, processing and device power requirements.

Scores use a 1–5 scale, where 5 is most suitable. Weighted score = Σ(weight × rating)/100. Ratings are engineering judgements based on the properties discussed in Q2–Q5, not universal rankings.

6.2 Ratings by deployment

Deployment	SDMA	TDMA	FDMA	CDMA	OFDMA	Recommended choice
Wide-area cellular voice	4	4	3	4	5	OFDMA with spatial beamforming; voice QoS scheduling.
Jam-resistant tactical link	2	3	2	5	3	CDMA / spread-spectrum primary, with frequency agility where permitted.
High-rate stadium downlink	4	3	2	3	5	OFDMA with SDMA/beamforming for spatial reuse.
5,000 low-rate sensors	2	4	3	4	3	Scheduled TDMA or hybrid random-access + TDMA; use compact reports and sleep cycles.

6.3 Example weighted scores

Deployment	SDMA	TDMA	FDMA	CDMA	OFDMA	Top selection
Wide-area cellular voice	3.85	3.70	3.05	3.85	4.45	OFDMA

Jam-resistant tactical link	2.80	3.35	2.70	4.45	3.30	CDMA
High-rate stadium downlink	4.10	3.45	2.95	3.45	4.55	OFDMA
5,000 low-rate sensors	2.90	4.25	3.45	3.85	3.75	TDMA

The table gives illustrative weighted outcomes using the stated criteria. In a real decision, each individual rating should be entered in a spreadsheet and calculated directly; scores should be updated if Q2–Q5 experiments show different measured performance.

6.4 Why systems combine techniques

Real systems combine access dimensions because they solve different problems. FDMA separates users into frequency bands, TDMA separates them in time, CDMA separates them by spreading codes, OFDMA allocates orthogonal subcarriers, and SDMA separates users spatially. A hybrid can therefore improve capacity and QoS—for example, OFDMA subcarriers can be scheduled in time and reused by directional beams.

6.5 Sensitivity check

Changing each criterion weight by ±20% and renormalising the weights is expected to leave the broad choices stable: OFDMA remains the strongest choice for high-rate stadium downlink; CDMA remains preferred for the tactical spread-spectrum case; and TDMA remains suitable for low-rate scheduled sensors. Wide-area voice is the least robust selection because OFDMA and CDMA can exchange rank depending on the importance assigned to robustness, latency and spectral efficiency.

6.6 Conclusion

No single multiple-access technique is best for all deployments. The correct selection depends on traffic shape, mobility, interference environment, latency, energy budget and user density. Hybrid designs are preferred when multiple constraints must be satisfied simultaneously.

Q7 – Diagnose, Redesign and Validate

Selected limitation: dedicated-channel allocation for bursty telemetry

The selected design is the dedicated FDMA/TDMA allocation considered in Q2. The limitation is resource reservation: each device holds a channel even though it transmits only a small report for a short interval. At low duty cycle, most reserved capacity is idle; when the number of devices exceeds the available channels, additional devices are blocked even though the average offered traffic is small.

A. Diagnose the bottleneck

- Capacity is consumed by channel reservations rather than by actual packet transmission time.
- The counted quantity that grows with device population is the number of reserved channels; it reaches 100 for 100 devices, while only 72 traffic resources are available.
- For 10% duty-cycle traffic, the average active fraction is only 0.10. Dedicated allocation therefore wastes about 90% of the assigned time when a device is silent.
- The original design assumes continuous or quasi-continuous traffic and does not exploit statistical multiplexing.

B. Design objective

Objective: serve all 100 telemetry devices using the same 72 usable traffic resources, avoid permanent channel reservation for silent devices, and deliver every 200-byte report within a defined 2-second burst window under the stated synchronised-arrival test. The design must not increase licensed bandwidth.

C. Revised strategy

Replace permanent per-device allocation with a shared scheduled TDMA pool. Devices retain their identity and queue reports locally. A lightweight request/grant mechanism or configured periodic report window informs the base station which devices have data. The scheduler assigns short transmission opportunities to active devices only. Unused opportunities are reassigned to queued devices.

Simplified procedure:

1. At each reporting interval, each device stores its report in a local queue.
2. Device sends a short access request or uses its configured contention opportunity.
3. Base station collects active-device requests and orders them by deadline/age.
4. Scheduler assigns available TDMA slots from the shared pool; each slot carries one report.
5. Devices acknowledge successful delivery; failed reports remain queued for retry.
6. At the end of the window, record delivered, delayed and dropped reports and update scheduling statistics.

Pseudocode:

```
FOR each reporting window:
  active_queue = collect_requests_from_devices()
  sort active_queue by deadline, then waiting_time
  slots = 72
  WHILE slots > 0 AND active_queue not empty:
     device = pop_oldest_or_earliest_deadline(active_queue)
     allocate_one_slot(device)
     transmit_and_acknowledge(device.report)
     slots = slots - 1
  carry_forward_unserved_reports(active_queue)
```

The redesign introduces request signalling, queue state and scheduler complexity. It reduces idle reservation and permanent blocking, but a synchronised burst can still create delay if the number of simultaneous reports exceeds the available slots in one scheduling window.

D. Prediction before implementation

For 100 synchronised reports and 72 available slots, dedicated allocation should admit at most 72 devices and block 28. The pooled scheduler should deliver 72 in the first slot batch and the remaining 28 in a second batch, so all 100 are delivered within two batches, assuming the second batch fits inside the 2-second window. Resource use is based on active reports rather than permanent reservations.

E. Implementation and validation

Measure	Original dedicated allocation	Shared scheduled TDMA redesign
Devices requesting service	100	100
Usable resources	72	72
Reports admitted in first batch	72	72
Reports permanently blocked	28	0
Reports carried to next batch	0 (blocked)	28
Total delivered within two batches	72	100
Resource reservation	One resource held per admitted device	Slots assigned only to active devices

Validation calculation: the original scheme has admission ratio 72/100 = 72% and blocking ratio 28/100 = 28%. The redesigned scheme has a two-batch delivery ratio of 100/100 = 100%, with 28% of reports delayed to the second batch rather than rejected. This is a deterministic workload calculation using the stated scheduler model; a code implementation should additionally test random arrivals, retransmissions, packet errors and different report sizes.

F. Defend the final design

The redesign meets the objective because it converts permanent reservation into demand-driven scheduling. In the synchronised 100-device case, it increases delivery from 72 to 100 reports over two batches, eliminating 28 permanent blocks. The trade-off is additional signalling and queueing delay. Dedicated allocation can still be preferable for continuous voice or strict low-latency traffic because it provides predictable access without request/grant overhead.

Overall conclusion

The seven questions show that mobile systems must be designed around traffic behaviour and operating constraints rather than the wireless medium alone. Offline application queues address intermittent connectivity; dynamic TDMA improves utilisation for bursty reports; CDMA requires synchronisation and power control; OFDMA balances throughput and fairness through scheduling; and SDMA improves reuse when users are spatially separable. A weighted comparison confirms that access techniques are complementary. For the selected telemetry

bottleneck, a shared scheduled TDMA pool is a practical redesign because it reduces idle reservations and converts blocking into controlled delay without increasing spectrum.

References

1. ITA03 – Mobile Computing, Assignment 2 – CO2, supplied question paper (8 pages).

2. Simon Haykin, Communication Systems, 5th Edition, Wiley – digital modulation and spread-spectrum communication concepts.

3. Theodore S. Rappaport, Wireless Communications: Principles and Practice, 2nd Edition, Prentice Hall – cellular radio, propagation and antenna-array concepts.

4. Andrea Goldsmith, Wireless Communications, Cambridge University Press – multiuser communications, OFDM and wireless channel models.

Note: Numerical overhead percentages, traffic models, SNR values, target rates and scheduling-window assumptions in this report are explicitly chosen modelling assumptions, not quoted standard parameters.